

Andrew Liu

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EDUCATION

University of California, Santa Barbara
Bachelor of Science, Electrical Engineering
GPA: 3.98

Expected Graduation: June 2028

RELEVANT COURSEWORK

Completed: Digital & Analog Circuits, Digital Design Principles, Electromagnetism, Photonics, Modern Physics, Circuits & Physics Labs, Linear Algebra, Differential Equations, Vector Calculus, Data Structures & Algorithms (C++)

In Progress: Signals & Systems, Solid State Electronics, VLSI, Fields & Waves

PROFESSIONAL EXPERIENCE

Electrical Engineering Intern

June 2026 – September 2026

Hoplynk, San Francisco, CA

- Designed, debugged, and integrated power distribution & battery monitoring printed circuit boards
- Designed and integrated power distribution systems using COTS electronics under time constraints
- Created write-ups and documents to explain design choices, allowing for smooth design reviews
- Assembled and tested various electronics using a multimeter and running test software (e.g., radios)

Avionics/Electrical Team Lead

January 2026 – Present

Gauche Rocket Project, Santa Barbara, CA

- Design data acquisition systems and boards to log and stream rocket telemetry during flight & tests
- Verify electronic components and sensor functionality with lab test equipment (multimeter, oscilloscope)
- Document test results and create protocols so the team has reference data and test instructions

PROJECTS

Power Distribution & Battery Monitoring Printed Circuit Board

June 2026 – September 2026

Hoplynk, San Francisco, CA

- Designed a power distribution & battery charging board with DC inputs (**12V–32V**) + smart charger
- Used MOSFETs at IC pins to ensure pack-level isolation during battery discharge & charge
- Optimized board size while also derating all components to below 50% of max voltage for reliability
- Ensured EMI from high-power switching areas did not affect analog readings and digital ICs
- Implemented a digital switch to allow cut-off of a heavy load, increasing system runtime by **~86%**
- Wrote and flashed STM32 firmware and Python scripts to send battery data and commands via USB

Sensor Logger Printed Circuit Board

April 2026 – May 2026

Gauche Rocket Project, Santa Barbara, CA

- Designed a 4-layer PCB in KiCad to log temperature, pressure, & acceleration data during rocket flight
- Selected integrated circuits for size, power, sensor compatibility, and avionics requirements
- Validated and tested key components and sensors (e.g., thermocouple amplifier) on breakout boards

Boost Converter Printed Circuit Board

March 2026 – April 2026

Personal Project

- Designed and prototyped a boost converter circuit, achieving an efficiency of **~66% (5V to 12V)**
- Implemented pulse width modulation control with closed-loop voltage feedback using Arduino/ATtiny
- Designed PCB layout, taking into account trace width calculations and component placement

SUMMARY OF SKILLS

Technical: Analog and Digital Circuit Analysis, Circuit Prototyping and Debugging, KiCad, PCB Design & Rework, Soldering, Power Electronics, LTspice, C++, Python, Oscilloscope, Multimeter, Function Generator